Natural Language to Database Query System (Text-to-SQL)

REPORT CONTENT

1. Introduction

This system enables non-technical users to query databases using natural language.

2. System Architecture

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User Input → Prompt + DB Schema → Groq LLM → SQL Query → Validation Layer → Database Execution → Output
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3. Key Components Used

- ChatGroq (LLM generation)
- PromptTemplate (schema-aware prompting)
- SQLite database engine
- SQL validation layer (security enforcement)

4. Challenges & Solutions

Challenge 1: Unsafe SQL queries

Solution: Blocked DROP/DELETE/UPDATE commands

Challenge 2: Ambiguous user input

Solution: Schema-aware prompting + assumptions

Challenge 3: Invalid SQL generation

Solution: Output cleaning + error handling

5. Use Cases

- Business analytics
- Data exploration
- Self-service reporting systems

6. Conclusion

The system demonstrates how LLMs can bridge the gap between natural language and structured databases.